Assignment 8 Solution

Q.1. What are the two main composition operations/strategies for fuzzy relations discussed in the lecture?

(A) Min-max composition and min-product composition

(B) Max-min composition and max-product composition

(C) Addition composition and multiplication composition

(D) Intersection composition and union composition

(E) None of the above

Solution: (B)

Q.2. What is the dimension of the resulting fuzzy relation matrix when performing max-min composition between a 3x4 matrix and a 4x2 matrix?

(A) 3x4

(B) 4x4

(C) 3x2

(D) 4x2

(E) None of the above

Answer: (C)

Q.3. What is the key difference between max-min composition and max-product composition of fuzzy relations?

(A) The order of matrices

(B) The operation used to combine membership values

(C) The universe of discourse

(D) The dimension of the resulting fuzzy relation matrix

(E) None of the above

Answer: (B)

Q.4. What does the weak distributivity over intersection property state?

(A) The composition of fuzzy relations is a subset of the intersection of compositions of fuzzy relations.

(B) The composition of fuzzy relations is equal to the intersection of compositions of fuzzy relations.

(C) The intersection of compositions of fuzzy relations is equal to the composition of fuzzy relations.

(D) None of the above

Answer: (A)

Q.5. Which property ensures that if one fuzzy relation is a subset of another, their compositions follow a similar relationship?

(A) Associativity

(B) Distributivity over union

(C) Monotonicity

(D) None of the above

Answer: (C)

Q.6. What will be the outcome if fuzzy relations R, S1, and S2 are composed according to the distributivity over union property?

(A) The composition of R and the union of S1 and S2 is equal to the union of compositions of R and S1 and R and S2.

(B) The composition of R and the union of S1 and S2 is equal to the composition of R and S1.

(C) The composition of R and the union of S1 and S2 is equal to the composition of R and S2.

(D) The composition of R and the union of S1 and S2 is equal to the composition of R and S1 and S2.

(E) None of the above

Answer: (A)

Q.7. Which of the following properties must be satisfied for a fuzzy relation set to be considered a fuzzy tolerance relation?

(A) Transitivity and asymmetry

(B) Symmetry and reflexivity

(C) Symmetry and transitivity

(D) Reflexivity and asymmetry

Answer: (B)

Q.8. What does the reflexivity property of a fuzzy tolerance relation entail?

(A) All elements in the fuzzy relation matrix are equal to 1

(B) All diagonal elements in the fuzzy relation matrix are equal to 1

(C) All off-diagonal elements in the fuzzy relation matrix are equal to 1

(D) All elements in the fuzzy relation matrix are non-zero

Answer: (B)

Q.9. In a fuzzy tolerance relation, what does the symmetry property indicate?

(A) All elements in the fuzzy relation matrix are symmetrically arranged

(B) Membership values are equal if the row and column indices are interchanged

(C) The fuzzy relation matrix is symmetric along the main diagonal

(D) Membership values are equal in all rows and columns

Answer: (B)

Q.10. What happens if a fuzzy relation set fails to satisfy the reflexivity property?

(A) It automatically satisfies the symmetry property

(B) It is considered a fuzzy equivalence relation

(C) It cannot be considered a fuzzy tolerance relation

(D) It must satisfy the transitivity property to compensate

Answer: (C)